Mphele Moswane

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SUMMARY

Final-year Software Engineering student specialising in backend systems and scalable APIs. Two-time hackathon champion: 1st Place nationally (ITWeb Security Summit 2026) and continentally (Mjanga AI Hackathon). Builds production-grade systems across Java, Go, Python, and TypeScript. CompTIA A+ certified.

SKILLS

Languages: Java, Python, Go, TypeScript, JavaScript
Frameworks: Spring Boot, FastAPI, Node.js, Express
Databases: PostgreSQL, MySQL, SQLite, MongoDB
Tools & Platforms: Git, GitHub, GitLab, Linux, VS Code, IntelliJ IDEA / JetBrains IDEs
Concepts: REST API design, OOP, Microservices, Event-driven architecture, ML integration

PROJECTS

Doria : AI Supply Chain Security Tool **1st Place National (IT web security summit)** - 2026
Rust · Python · TypeScript / Node.js · VS Code Extension API | github.com/doria-sec · Demo

- Architected a zero-trust package interception engine using a local forward-proxy model that silently intercepts npm install workflows via automated .npmrc workspace injection, with zero developer configuration and no elevated system privileges.
- Eliminated cache-staleness supply chain attacks by enforcing a multi-key registry validation schema indexed by Name + Version + Cryptographic Hash (SHA), ensuring any upstream code mutation triggers a mandatory re-scan.
- Built performance-critical VS Code extension middleware using in-memory workspace caching and background child-processes to stream multi-threaded threat telemetry as real-time visual diagnostics without editor lag.
- Led technical design and execution that secured 1st Place National Champions out of all competing teams at the ITWeb Security Summit Hackathon 2026.

AfyaAI : Multilingual Emergency Medical Assistant **1st Place Continental (Mjanga AI hackathon)** - 2026
Python · FastAPI · Uvicorn · Llama 3.1 · Google Cloud TTS · Deep Translator | github.com/elviscgn/AfyaAI

- Architected a high-concurrency asynchronous FastAPI backend using run_in_threadpool to offload TTS generation, translation, and LLM inference to worker threads, keeping the ASGI event loop fully responsive under simultaneous load.
- Engineered custom LLM prompting guardrails for Llama 3.1 that forced parseable severity metadata output (e.g. [SEVERITY: HIGH]), enabling the backend to programmatically trigger emergency geolocation routing to nearest physical clinics.
- Built a fault-tolerant TTS pipeline with graceful language fallback logic, intercepting unavailable regional dialects (Zulu, Swahili, Afrikaans, Northern Sotho) and safely degrading to stable voice models without crashing the server.
- Won 1st Place at the Africa-wide Mjanga AI Hackathon, beating teams across the continent.

Uwatu: Event-Driven Agritech IoT Platform - 2026
Go · Python · FastAPI · Pydantic v2 · Google Gemini · Nokia Network as Code API | github.com/Uwatu · Demo

- Engineered a hybrid event classification engine with three processing layers: deterministic reflex rules for instant threat flagging, Welford's online algorithm for O(1) real-time Z-score behavioural baselines, and an Isolation Forest ML model for multivariate anomaly detection.
- Integrated the Google Gemini SDK to power a contextual alerting narrator that translates raw telemetry and ML confidence scores into actionable SMS alerts capped at 160 characters for low-bandwidth cellular delivery.
- Bridged a high-concurrency Go ingestion backend with the Python predictive layer, routing Nokia Network as Code API telecom signals to correlate network anomalies with physical livestock threats.

EDUCATION

WeThinkCode_ Sept 2025 – Dec 2026
Software Development Diploma

- Built a solid foundation across Python, Java, Go, and JavaScript through a rigorous, project-based curriculum covering OOP, data structures, and software architecture.
- Practised professional engineering workflows including TDD, CI/CD pipelines, Docker containerisation, and SQL-based persistence with ORMs.
- Developed client-server and multi-threaded networked applications using sockets, and collaborated in agile teams through daily stand-ups, sprint planning, and retrospectives.

CERTIFICATIONS

CompTIA A+

VOLUNTEER EXPERIENCE

Peer Tutor - Sept 2025 – Present
*WeThinkCode_*

- Support first-year students during Code Clinics, helping them debug programs, grasp core concepts, and apply test-driven development practices.
- Review peers' code submissions and provide constructive feedback to reinforce best practices in code quality, git workflow, and object-oriented design.

AWARDS

1st Place National Champions ITWeb Security Summit Hackathon 2026 | *Doria · Sandton Convention Centre, Johannesburg*
1st Place Continental Mjanga AI Hackathon 2026 | *AfyaAI · Africa-wide competition*